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Comparative analysis of human Mesenchymal Stromal Cells from Adipose Tissue and Dental Pulp: phenotypic characterization and secretome profiling

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Abstract

Background In tissue regeneration, as well as in post-traumatic recovery or in treating pathological alterations, mesenchymal stromal cells (MSCs) and their products for cell-free treatments are increasingly attractive and applicable. For this reason, there is an urgent need to thoroughly investigate MSCs of different origins, especially those readily available and with no ethical concerns obtained from healthy donors.

Methods Human MSCs were derived from discarded adipose tissue of four donors (ADSCs; 8 cell populations isolated by enzymatic digestion and mechanical fragmentation) and dental pulp of two donors (DPSCs; 4 cell populations from radicular and coronal compartments by mechanical fragmentation). Cells were characterized by differentiation, proliferation, and morphological features. Conditioned media (CM) were collected, and the secretome profile analyzed.

Results The trilineage differentiation assay and CD immunophenotyping showed that all primary cell lines possessed typical MSC characteristics, apart from the inability of DPSCs to perform adipogenesis. Significant CD differences were found mainly due to tissue source and regional compartments regarding coronal vs. radicular dental pulp. Notably, DPSCs were consistently smaller, Nestin-positive, and had a higher proliferation rate than ADSCs. Secretome analysis regarding anti-inflammatory and pro-inflammatory cytokines, chemokines, and growth factors accumulating in the CM throughout the culture showed significant variations among MSC lines from the two tissues and within ADSCs obtained with different extraction methods. All MSC populations release a comparable number of extracellular vesicles (EVs), although ADSCs appeared to produce a significantly higher number of smaller exosomes than DPSCs.

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Depending on the tissue of origin, MSCs released specific sets of microRNAs, either free or enclosed in EVs, impacting many cellular processes. The microRNAs more expressed from DPSCs are mainly involved in oxidative stress and apoptosis pathways, while those of ADSCs play a regulatory role in cell cycle and proliferation.

Conclusions The results support the notion that, despite their common characteristics, MSCs can differ in many aspects related to their ontogeny, extraction method, and, to a lesser extent, regionalization and donor heterogeneity. These findings pose challenges for the clinical translation of MSCs, their CMs, and derivatives and underline the importance of standardizing protocols to obtain MSC products from their secretome.

Keywords Human mesenchymal stromal cells, Adipose tissue, Dental pulp, Secretome, Conditioned medium, Extracellular vesicles, miRNAs

Introduction

Mesenchymal stromal cells (MSCs) are spindle-shaped plastic-adherent cells isolated from several adult and fetal tissues such as bone marrow, adipose tissue, amniotic fluid, umbilical cord, and Wharton's jelly [1], with multipotent differentiation capacity in vitro [2–3]. Increasing evidence demonstrates that, besides some common characteristics, MSCs exhibit different biological properties depending on the tissue's anatomical localization and the donor's age and health status. Several clinical trials at various phases demonstrate the safe and efficient use of MSCs in treating autoimmune, inflammatory, and degenerative diseases (<https://www.clinicaltrials.gov/>). From thriving research, it is emerging that such beneficial effects are mainly due to the MSC's ability to secrete a broad range of bioactive molecules with regenerative impact on injured tissues rather than directly replacing damaged cells [4–6]. Supporting such a notion, in vitro and in vivo studies on animal models demonstrated that MSC conditioned media (CMs) can significantly reduce tissue damage in injured heart, lung, brain, cornea, and musculoskeletal tissues, among others, and stimulate their regeneration [7–11]. MSCs can produce many soluble factors (including cytokines and growth factors) in vitro with proliferative, anti-apoptotic, and immunomodulatory effects [12–15]. MSCs produce and release such compounds, defined secretome, either in soluble form and enclosed in extracellular vesicles (EVs) as microvesicles and exosomes [16–19]. Furthermore, it is essential to note that MSC secretome composition differs for the type and quantity of compounds [20–22]. These differences are crucial for implementing cell-free therapy, which makes the study of MSCs' secretomes essential to know how to produce the most suitable CM for treating a given disease.

The present study focused on characterizing human MSCs derived from adult adipose tissue (AT) and dental pulp (DP) and evaluating the content of their secretomes. These cells differ from others in that the tissues they reside in are easily accessible, requiring minimal invasive surgery, and are easy to culture after isolation [23–29]. In particular, we obtained primary cell lines from adipose

tissues termed adipose derived stromal cells (ADSCs) isolated by two different methods: the Stromal Vascular Fraction based on enzymatic digestion (SVF) and the explant method based on Mechanical Fragmentation (MF) [30, 31], and primary lines of DP-derived stromal cells (DPSCs) obtained from open apex third molars: the radicular compartment (RPSCs) and the coronal compartment (CPSCs) of dental pulp, both extracted by MF [32].

The results show that the adipose and dental pulp primary cell populations present differences in proliferation and differentiation, together with immunophenotyping, morphological, cytoskeletal, and secretome profiling. This implies that studies are needed to fully characterize the MSCs and their derivatives to be applied for more personalized therapeutic interventions.

Materials and methods

Isolation and culture of ADSCs

Human ADSCs were obtained from abdominal adipose tissue harvested from four healthy donors aged 45–60 years (female samples #3 and #4, and male samples #5 and #6) at the Department of Orthopedics at the Policlinico Tor Vergata, according to the policies approved by Ethical Committee of Fondazione PTV Policlinico Tor Vergata with authorization number 160/20.

The tissue was immediately processed by using the LIPOGEMS® system [30, 31], a disposable kit that separates the lipoaspirate into three layers: an upper layer of oil derived from the lysis of adipocytes, an intermediate layer of intact adipose tissue, and a lower layer containing cellular contaminants such as erythrocytes. Samples, kept at 4°C, were then transferred to the laboratory and further processed. The top and bottom layers were initially aspirated and discarded, whereas the middle layer (LG) was utilized following two different isolation methods. In the first, called Mechanical Fragmentation (MF), half of each LG fraction was put in α MEM (Aurogene, Rome, Italy) supplemented with 2 mM L-glutamine, 100 IU/ml penicillin, 0.1 mg/ml streptomycin (all from Sigma-Aldrich, Milan, Italy) from here on called Basic Medium (BM), plus 20% FBS (Gibco, Life Technologies,

Paisley, Scotland). Cultures were carried out in cell culture dishes in a humidified incubator at 37°C in 5% CO₂ in humidified air. Over 2 weeks, ADSCs (termed ADSCs-MF) outgrew from the fragments floating on the medium surface, adhered to the bottom of the plate, and proliferated. When approximately 80% confluence was reached, the cells were detached from the plate by incubation with trypsin-EDTA (Sigma-Aldrich) for 5' at 37°C, re-plated at 1×10^3 cells/cm² for expansion in BM supplemented with 10% FBS (Gibco) and used for experiments at 4th-6th passage. The remaining part of each LG fraction was extracted using the second method called Stromal Vascular Fraction (SVF), as previously described [33]. In this procedure, the LG fraction was washed twice with Dulbecco's phosphate-buffered saline (DPBS; Gibco) and subjected to overnight enzymatic digestion at 37°C with collagenase 1A (Sigma-Aldrich). The digested material was then centrifuged (10' at 1200 g) and the cell pellet plated onto tissue culture dishes, in BM supplemented with 10% FBS (Gibco). At approximately 80% confluence, the cells were detached by incubation with trypsin-EDTA (Sigma-Aldrich) for 5' at 37 °C and re-seeded at 1×10^3 cells/cm² density. These cells, named ADSCs-SVF, were subcultured and used for experiments in the 4th-6th passage.

Isolation and culture of DPSCs

Human DPSCs were obtained from sound third molar teeth with open apex from two healthy donors aged 18–20 years (female samples #1 and #2) at the Dental Clinic of Policlinico Tor Vergata, according to the policies approved by the Ethical Committee of Fondazione PTV Policlinico Tor Vergata with authorization number 172/22. After extraction, teeth were transferred to the laboratory in DPBS at 4 °C, washed several times, and pulps were isolated into the same medium following the explant method of fragmented tissue as previously described [32], with some modifications. Briefly, each tooth was cut at the amelo-cement junction with a diamond disc connected to a micro motor, and the pulp was gently removed using a sterile dental scalpel. Each pulp was then cut with a 15-blade scalpel into two portions, the coronal and the radicular, and each portion further fragmented into small pieces of 1–2 mm³ that, after washing by centrifugation at 300 g for 5', were seeded onto tissue culture plates in BM supplemented with 10% FBS (Gibco). Cultures were carried out inside a humidified incubator at 37 °C with 5% CO₂ in air. After two to four weeks from seeding, cells outgrowing from the fragments began to form a monolayer and reached 80% confluence. At this time, cells were detached from the plate by incubation with trypsin-EDTA (Sigma-Aldrich) for 5 min at 37 °C in humidified air and re-plated at 2.5×10^3 cells/cm² in BM supplemented with 10% FBS (Gibco).

The cells were then sub-cultured and used for experiments at the 4th to 6th passage. Coronal Pulp Stromal Cells (CPSCs) and Radicular Pulp Stromal Cells (RPSCs) were obtained from the coronal and radicular pulp.

Osteogenic, chondrogenic, and adipogenic differentiation

Trilineage differentiation of MSCs was performed following established protocols [33]. For osteogenic induction, cells were seeded at 3×10^3 cells/well in 48-well plates and cultured in BM supplemented with 10% FBS (Gibco), at 37 °C and 5% CO₂ in humidified air.

After 16 hrs, the medium was replaced with the osteogenic differentiation medium: DMEM supplemented with 10% FBS (Gibco), 2 mM L-glutamine, 100 IU/ml penicillin, 0.1 mg/ml streptomycin, 50 µM ascorbic acid-2 phosphate, 10 mM β-glycerophosphate, and 0.1 µM dexamethasone (all from Sigma-Aldrich). The medium was changed every 3–4 days for 21 days to achieve osteogenic differentiation, and the osteogenic phenotype was subsequently confirmed by Alizarin Red staining. Briefly, the cells were washed with PBS and fixed with 4% (w/v) paraformaldehyde (PFA) for 15' at room temperature (rt). Then, staining was performed with 40 mM Alizarin Red S (from Sigma-Aldrich) for 20', washed with H₂O, and finally air-dried at rt. Mineralized matrix deposition appeared red under light microscope observation.

The chondrogenic differentiation was performed using the micro mass culture technique. Briefly, 1×10^5 cells suspended in 10 µl BM supplemented with 10% FBS (Gibco) were plated in the center of the well of a 48-well plate and allowed to adhere for 2 h, and then 200 µL of BM supplemented with 10% FBS (Gibco) was added. After 16 h from cell seeding, the medium was replaced with complete chondrogenic differentiation medium: DMEM supplemented with 10% FBS (Gibco), 2 mM L-glutamine, 100 IU/ml penicillin, 0.1 mg/ml streptomycin, 6.25 µg/ml human insulin, 50 nM ascorbic acid-2 phosphate (all from Sigma-Aldrich) and 10 ng/ml TGF-β1 (Peprotech). The medium was changed every 3–4 days for 21 days. At the end of the differentiation period, the spheroids were fixed in 4% PFA for 24 h, dehydrated in ascending concentrations of ethanol, cleared in xylene, embedded in paraffin, cut into 5 µm-thick sections, and finally stained with Alcian Blue 8GX.

To induce adipogenic differentiation, DPSCs and ADSCs were seeded, respectively, at 4×10^4 and 3×10^4 cells/well in 48-well plates and cultured in BM supplemented with 10% FBS (Gibco), at 37°C and 5% CO₂ in humidified air. After 16 hrs, the medium was replaced with the adipogenic differentiation medium: DMEM supplemented with 10% FBS (Gibco), 2 mM L-glutamine, 100 IU/ml penicillin, 0.1 mg/ml streptomycin, 0.5 µM 3-isobutyl-1-methylxanthine, 50 µM indomethacin, and

0.5 μ M dexamethasone (all from Sigma-Aldrich). The culture medium was replaced with fresh medium every 3–4 days for 21 days, and the adipocyte differentiation was confirmed by Oil Red O staining. Briefly, cells were washed with PBS twice and fixed with 4% PFA for 15' at rt. Then, cells were covered with 0.2% Oil Red O working solution (from Sigma-Aldrich) for 30' at rt and washed with PBS. The cytoplasmic lipid droplets appeared red upon light microscope observation.

For the negative control of osteogenic, chondrogenic, and adipogenic differentiation, cells were cultured in DMEM supplemented with 10% FBS without differentiation factors or inducers and stained at the end of the culture time in parallel with the treated samples.

Flow cytometry (FC) analysis

FC detected the expression of mesenchymal markers on DPSCs and ADSCs. Sub-confluent, adherent DPSCs and ADSCs were detached by 0.25% trypsin/EDTA (Gibco), washed, and suspended in PBS containing 2 mM EDTA and 1% FBS (Running Buffer). Cells were then stained in 30 μ L of antibodies diluted in Brilliant Stain Buffer (BD Biosciences, Franklin Lakes, NJ) for 20' at rt, as described [34]. After two washes in Running Buffer, the cell suspensions were immediately acquired on a daily calibrated Cytoflex LX flow cytometer. Flow Cytometry Standard (FCS) files were analyzed with FlowJo (BD Biosciences). Spillover subtraction (compensation) was done using single-fluorochrome-stained VersaComp Beads (Beckman Coulter, Miami, FL). Table S1 (in Additional file 1) shows the complete list of FC reagents used. Principal Component Analysis (PCA) and Hierarchical Clustering (HC) were performed by ClustVis [35].

Immunofluorescence analysis

For immunofluorescence (IF), the culture medium was removed, and after 2–3 washes in DPBS, cells were fixed in 4% PFA for 15' at rt. Cells were permeabilized with 0.1% Triton X-100 in PBS for 10', blocked in 3% BSA for 30', and then incubated with primary antibodies overnight at 4 °C. The primary antibodies used were anti-Nestin (ab-105389, Abcam, Cambridge, UK) (1:500), anti-Vimentin (ab-45939, Abcam) (1:300), and anti- α SMA (A2547, Sigma-Aldrich) (1:300). The secondary antibodies used for IF were Alexa Fluor 488-labeled goat anti-mouse (Thermo Fisher Scientific, Milan, Italy) (1:500) and Alexa Fluor 488-labeled goat anti-rabbit (Thermo Fisher Scientific) (1:500), all incubated for 1 h at rt. Hoechst was added to stain cell nuclei. Cells were mounted in PBS-glycerol (1:1) and analyzed by a Leica DMI6000B microscope. To quantify different cell populations in culture, at least 150 cells within a randomly selected field were scored to reach 500 cells in at least three independent experiments.

Water-soluble tetrazolium salt (WST-1) cell proliferation assay and doubling time analysis

ADSCs and DPSCs were seeded at 800 cells/well in 96-well plates, with four duplicate wells in each group. Cell proliferation was evaluated after 24, 48, 72, 96, and 168 h of culture. At the end of the experimental period, the water-soluble tetrazolium salt (WST-1) assay (05015944001, Roche) was performed following the manufacturer's instructions. In brief, cells were washed twice with DPBS and incubated in fresh medium with 10% WST-1 reagent for 2 h. Absorbance was determined in a microplate reader (Infinite F50, Tecan) at 450 nm (600 nm was used as reference wavelength and subtracted). Data from four wells containing culture medium plus WST-1 in the absence of cells were used as background control (blank) for the ELISA reader.

Doubling time was calculated from WST-1 absorbance data during the proliferative phase (up to 96 h). Nonlinear regression curve fitting was applied with the exponential growth model using GraphPad Prism software v 8.0 (La Jolla, CA, USA).

Collection of conditioned medium

MSCs at 4th passage were seeded as described above and cultured to 70–80% confluence in BM supplemented with 10% FBS (Gibco). Then, the cells were washed 3 times with 10 mL of DPBS (Gibco) without Ca^{2+} and Mg^{2+} and placed in serum-free BM at 37°C, with 5% CO_2 in humidified air. Cells were cultured for 3 days (3d). Then, CM samples were collected and centrifuged at 200 g for 15' at 4°C to remove eventual dead cells. The supernatant was centrifuged at 780 g for 10' at 4 °C to remove cellular debris, filtered, and frozen in aliquots at –80 °C.

In some experiments, two conditioning rounds were performed. Briefly, after the first round of conditioning for 4 days, the collected CM was used as medium for a new culture plate with freshly prepared cells at 70–80% confluence. After three additional days of conditioning (7 days in total), the obtained medium enriched with the molecules secreted from the second culture was named Concentrated Conditioned Medium (CCM) and was collected, processed, and stored as above.

Secreted cytokines, chemokines, and growth factors analysis

The targeted quantitative analysis of secreted factors in conditioned media (CM and CCM) collected as described above was performed using the Bio-Plex Pro Human Cytokine 27-plex Assay (#M500KCAF0Y, Bio-Rad, Milan, Italy) based on xMAP technology [36]. Magnetic beads labeled with red and infrared fluorophores are coated with specific antibodies, thus allowing the simultaneous detection of multiple target analytes within one sample. Following the reaction of beads with target

analytes, detection is performed with a biotinylated antibody and phycoerythrin-conjugated streptavidin. All steps were performed according to the manufacturer's instructions. The concentration of the following analytes was detected simultaneously within CM and CCM of RPSCs, CPSCs, ADSCs-SVF and ADSCs-MF: interleukin (IL)-1 β , IL-1 α , IL-2, IL-4, IL-5, IL-6, IL-7, IL-8, IL-9, IL-10, IL-12 (p70), IL-13, IL-15, IL-17, Interferon γ -induced protein 10 (IP-10), Eotaxin, Granulocyte-colony stimulating factor (G-CSF), Granulocyte macrophage colony stimulating factor (GM-CSF), Interferon (IFN) γ , Monocyte chemoattractant protein 1 (MCAF/MCP1), Macrophage inflammatory protein 1-alpha and beta (MIP-1 α and MIP-1 β), RANTES, Tumour necrosis factor alpha (TNF- α), Fibroblast growth factor (bFGF), Platelet-derived growth factor-BB (PDGF-BB), Vascular endothelial growth factor (VEGF). Data were acquired using a Bio-Plex MAGPIX Multiplex Reader system (Bio-Rad). Standard curve optimization and the calculation of analyte concentrations were performed using the Bio-Plex Manager software. Cytokine concentration was expressed in pg/ml. Measurements were performed in triplicate on 50 μ L of CM or CCM samples. Aliquots of BM (50 μ L) were also analyzed as negative controls. Data were expressed as mean $\pm$ SEM. Ordinary one-way ANOVA and Bonferroni post-hoc analysis were used to assess the significance between samples obtained from all the cell populations and between CM and CCM samples using the GraphPad Prism software v 8.0 (La Jolla, CA, USA).

EVs isolation and analysis

EVs were isolated from collected CM and CCM using ExoQuick-TC (System Biosciences, Palo Alto, USA). Briefly, supernatants were gently mixed with the ExoQuick-TC Precipitation Solution and incubated for 16 h at 4 $^{\circ}$ C. After incubation, samples were centrifuged at 1500 g for 30 min at 4 $^{\circ}$ C. The pellet containing EVs was suspended in 200 μ L of sterile DPBS (Gibco). Vesicle suspensions were examined by FC, as previously described [37], using a CytoFLEX S (Beckman Coulter Life Sciences, Brea, CA, USA). As recommended by the manufacturer, EV size ranges were set using the Gigamix beads solution, previously obtained by combining Megamix-Plus FSC reagent (BioCytex, Marseille, France), consisting of 100 nm, 300 nm, 500 nm, and 900 nm fluorescent beads, with Megamix-Plus SSC reagent (BioCytex) composed of 160 nm, 200 nm, 240 nm, and 500 nm fluorescent beads. CytoFLEX Daily Quality Control (QC) was utilized before each analysis to verify the optical alignment and the fluidics system integrity of CytoFLEX. To avoid any swarm detection phenomenon, EV samples dissolved in PBS were analyzed by FC, performing a serial dilution assay. Before each analysis, ddH₂O was

used to clean the fluidics system until we reached a maximum background noise of 150 events/s.

microRNA (miRNA) isolation and analysis

Total RNA was extracted using a commercial column purification system (miRNeasy Mini Kit, Qiagen) and on-column DNase treatment (RNase-free DNase Set, Qiagen). Library preparation has been carried out using the QIAseq miRNA library kit (Qiagen, Hilden, Germany). All samples have been sequenced in 75 single-end reads on the Illumina NextSeq500 platform. The quality control of raw sequencing reads was performed using FastQC v0.11.9 and MultiQC v1.10.1. The trimming of reads was performed with Cutadapt v3.4. The alignment on the miRbase human hairpin database release 22.1 was performed using SHRiMP v2.2.3. The BEDTools suite v.2.30.0 was used to select overlapping mappings on miRNA hairpins with mature miRNAs. A custom script was used for deduplication before counting mapped reads on mature human miRNAs (2656 miRNAs as annotated in the miRbase). For the analysis of miRNA expression profiles, the miRNA expression counts were imported into DESeq2 (v1.32.0) R package, using the median of ratios method (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4302049/>) for normalization and differential expression analysis (Wald test). The differentially expressed miRNAs for each comparison were selected using an adjusted p-value < 0.05. The Venn diagram was generated using Venny (<https://bioinfogp.cnb.csic.es/tools/venny/>).

To perform functional enrichment analysis, we uploaded the genes deregulated by the top twenty differentially expressed miRNAs between CM-DPSC and CM-ADSC to the Enrichr Web Server (<https://maayanlab.cloud/Enrichr/>), which allows us to classify the proteins based on the MSigDB Hallmark 2020 ontology. We performed separate pathway analyses for the genes/proteins deregulated by the first ten miRNAs upregulated in CM-DPSC and the first ten in CM-ADSC. The target genes for microRNAs were obtained using tarbase (<https://dianalab.e-ce.uth.gr/tarbasev9>).

To validate the expression of miRNAs, we chose 16 miRNAs (six upregulated in ADSCs, six upregulated in DPSCs, and four non-differentially expressed between the two CMs). We checked for their presence in freshly prepared CMs. Total RNA was extracted using a commercial column purification system (miRNeasy Mini Kit, Qiagen) with on-column DNase treatment (RNase-free DNase Set, Qiagen). Following the manufacturer's protocol, the first-strand complementary DNA (cDNA) was synthesized from 3 μ L of total RNA using the miRCURY LNA RT kit (Qiagen). Quantitative analysis of differentially and non-differentially expressed secreted miRNAs between ADSCs and DPSCs was performed

using real-time PCR. The miRCURY LNA SYBR Green PCR Kit (Qiagen) and a pre-spotted plate containing primers specific for the selected miRNAs (miRCURY LNA miRNA Custom PCR Panels, Qiagen) were utilized for this analysis. To ensure the quality of RNA isolation, cDNA synthesis, and PCR amplification, the RNA Spike-In Kit (Qiagen) was incorporated as a quality control measure. The expression values were normalized relative to the mean expression levels of hsa-miR-125b-5p and hsa-miR-3665, which were used as housekeepers. The geNorm software was employed to identify suitable endogenous miRNAs for data normalization. The ΔCT method was used to calculate the relative fold change of miRNA expression, and the values were expressed as $2^{-\Delta CT}$.

Statistical analysis

Statistical analyses are detailed in the Figure legends and in the corresponding Materials and Methods subsections describing each technique.

Results

As reported in Materials and Methods, unless otherwise indicated in the figures' notations, the results were obtained using twelve cell populations of human MSCs: eight from adipose tissue samples and four from dental pulp samples.

Trilineage differentiation assay of MSCs

To assess trilineage differentiation (adipogenic, chondrogenic, and osteogenic), the putative MSCs obtained from human adipose tissue (ADSCs) and dental pulp (DPSCs) were cultured according to the protocols described in Materials and Methods.

As shown in Fig. 1A-C, both DPSCs and ADSCs succeeded in differentiating toward osteogenic and chondrogenic lineages. Specifically, osteogenic differentiation was verified by mineralized matrix deposition through Alizarin Red staining (Fig. 1A). Upon chondrogenic differentiation, we found similar Alcian Blue staining of 3D culture in both DPSCs and ADSCs, confirming the formation of the specific extracellular matrix (Fig. 1B). In the negative control samples, none of these characteristics were observed. Following adipocyte differentiation, ADSCs

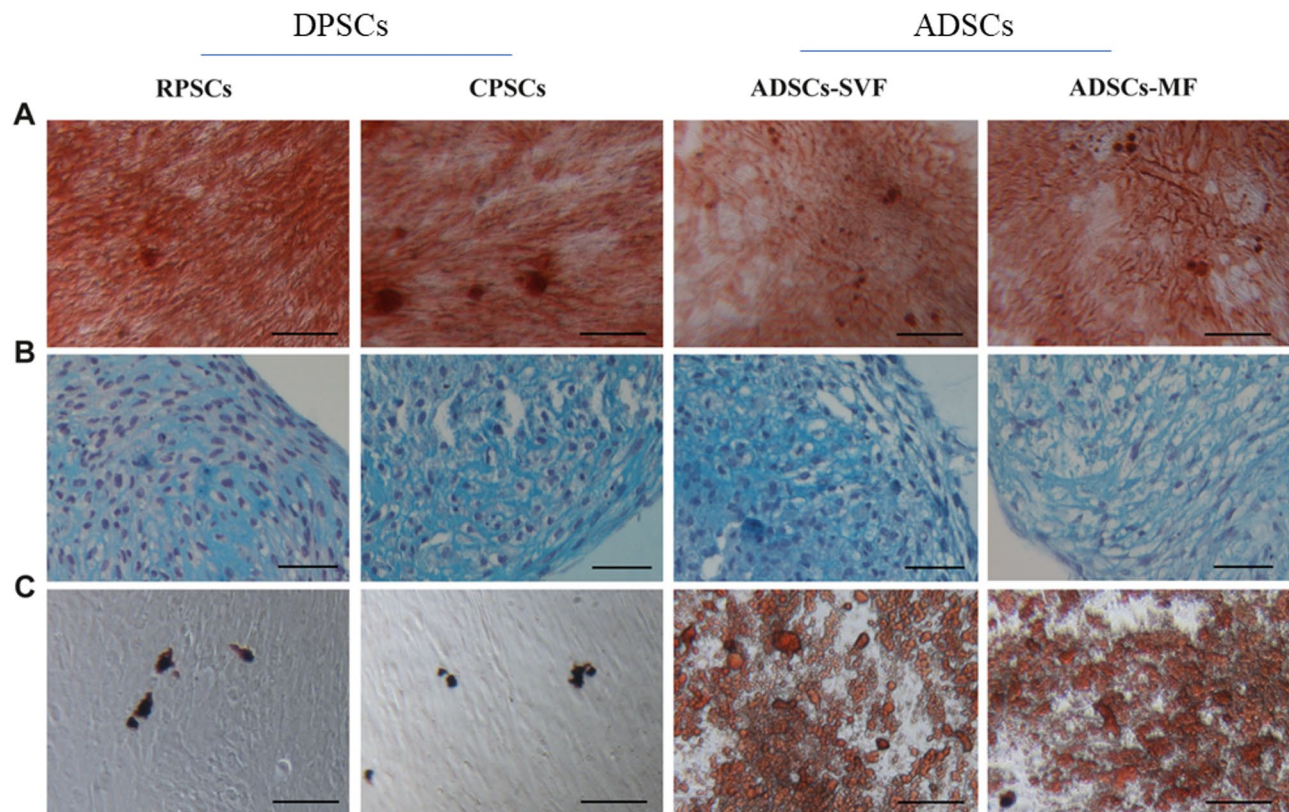


Fig. 1 Differentiation potential of dental pulp-derived stromal cells (DPSCs) and adipose-derived stromal cells (ADSCs). **(A)** Representative micrographs of osteogenic differentiation of DPSCs and ADSCs demonstrated by Alizarin Red staining. Scale bar = 100 μ m. **(B)** Representative micrographs of chondrogenic differentiation of DPSCs and ADSCs demonstrated by Alcian Blue staining. Scale bar = 50 μ m. **(C)** Representative micrographs of adipogenic differentiation of ADSCs but not DPSCs detected by Oil Red O staining. Scale bar = 100 μ m. DPSCs are obtained from the radicular compartment (RPSCs) and the coronal compartment (CPSCs); ADSCs are obtained by enzymatic extraction (SVF) and mechanical fragmentation (MF)

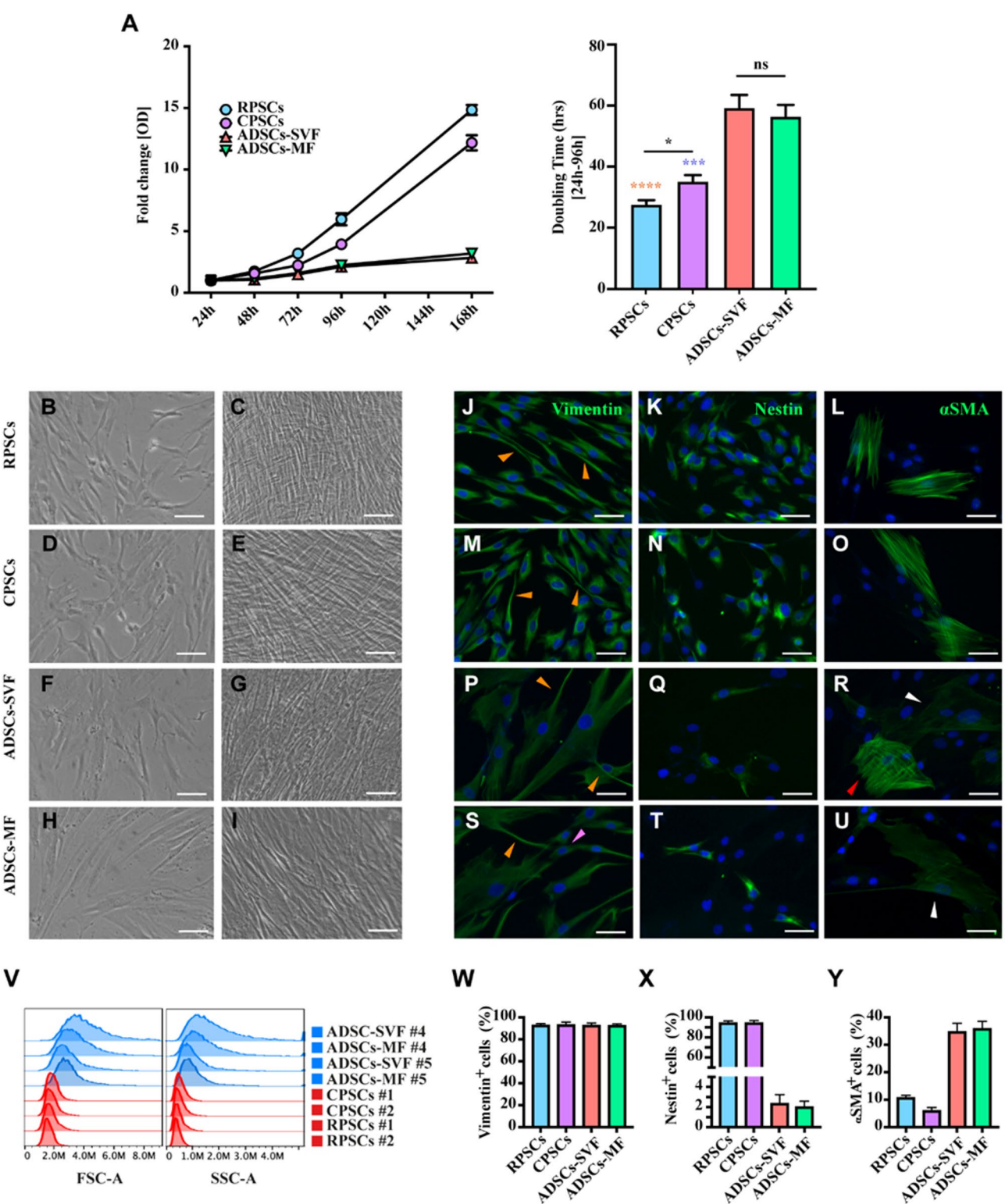


Fig. 2 (See legend on next page.)

differentiated into adipocyte-like cells, while DPSCs cultured under the same differentiation condition did not show cytoplasmic lipid droplets positive for the Oil Red O staining (Fig. 1C).

Proliferative, morphological, and cytoskeletal features of MSCs

The proliferative capacity of the ADSCs and DPSCs was estimated throughout one week of culture by WST-1

(See figure on previous page.)

Fig. 2 Proliferation and morphological analyses of DPSCs and ADSCs. **(A, left panel)** Measurement of growth of the DPSCs (RPSCs and CPSCs) and ADSCs (SVF and MF) throughout a week of culture. **(A, right panel)** Doubling time of such cells during the first 96 h of culture. Data are represented as mean $\pm$ SEM of three independent experiments. Results of one-way Anova test between the indicated groups (black asterisks), RPSCs vs. ADSCs-SVF/MF (orange asterisks), and CPSCs vs. ADSCs-SVF/MF (blue asterisks) are depicted. ns = not significant, * $p < 0.05$, *** $p < 0.001$, **** $p < 0.0001$. **(B–I)** Representative micrographs of RPSCs **(B, C)**, CPSCs **(D, E)**, ADSCs-SVF **(F, G)**, and ADSCs-MF **(H, I)** in sub-confluent **(B, D, F, H)** and over-confluent **(C, E, G, I)** cultures. Scale bar = 100 μ m. **(J–U)** Representative micrographs of immunofluorescence analysis of cytoskeletal Vimentin **(J, M, P, S)**, Nestin **(K, N, Q, T)**, and α SMA **(L, O, R, U)** performed on RPSCs **(J–L)**, CPSCs **(M–O)**, ADSCs-SVF **(P–R)**, and ADSCs-MF **(S–U)**. Orange arrowheads in J, M, P, and S indicate cytoplasmic extensions. White arrowheads in R and U indicate low levels of α SMA fluorescence. The red arrowhead in R indicates a high level of α SMA fluorescence. Scale bar = 50 μ m. **(V)** Cytometric analysis of cell dimension (forward scatter, FSC-A) and cell granularity (side scatter, SSC-A) of the MSC cell populations, as listed, two for each donor, based on extraction method (#4 and #5) or tissue localization (#1 and #2). In blue, the adipose-derived and in red, the dental pulp-derived. **(W–Y)** Graphs report the quantification of Vimentin **(W)**, Nestin **(X)**, and α SMA **(Y)** as a percentage of positive cells performed in the eight MSC cell populations, the same indicated in panel V. Data are expressed as Mean $\pm$ SEM

assay. The results showed that both DPSC populations were significantly more proliferative with respect to ADSCs and that, within DPSCs, RP grew faster than CP (Fig. 2A, left panel). Interestingly, considering the entire analyzed culture time, we observed that ADSCs nearly stopped proliferating after reaching 100% confluence (96–120 h of culture). In contrast, DPSCs continued to grow (Fig. 2A, left panel), forming multiple cell layers (Fig. 2C, E). The WST-1 data of the first 96 h of cell culture were utilized to obtain doubling time data, as indicated in the Materials and Methods section. ADSCs showed a significantly higher doubling time compared with DPSCs (RPSCs vs. ADSCs-SVF/MF $p < 0.0001$; CPSCs vs. ADSCs-SVF/MF $p < 0.001$) while there were no significant differences between the two ADSC cell populations (doubling time: ADSCs-SVF = 59.34 ± 4.16 h; ADSCs-MF = 56.48 ± 3.75 h) (Fig. 2A, right panel). In addition, RPSCs showed a lower doubling time in comparison to CPSCs (RPSCs = 27.72 ± 1.34 h vs. CPSCs = 35.26 ± 2.00 h, $p < 0.05$) (Fig. 2A, right panel).

ADSCs and DPSCs were analyzed under Phase Contrast Microscopy (PH) for their morphology and by IF for cytoskeleton components. Although most of the cells of all samples showed fibroblastic-like morphologies (Fig. 2B–I), some differences were observed. ADSCs showed a flatter and larger shape, as evidenced also by cytometric analysis (Fig. 2V, left panel) and Vimentin immunolocalization (Fig. 2J, M, P, S) that stain more than 90% of both ADSCs and DPSCs (Fig. 2W). Moreover, ADSCs presented fewer long and thin cytoplasmic extensions than DPSCs (Fig. 2J, M, P, S). Relevantly, >90% of both DPSC types were Nestin⁺ in comparison to only >1% of ADSCs⁺ (Fig. 2K, N, Q, T, X), while 5–10% DPSCs and about 30% of ADSCs resulted positively stained for α SMA (Fig. 2L, O, R, U, Y). These latter included ADSC-SVF with an equal proportion of high and low positive cells (Fig. 2R) and ADSC-MF, all with very low staining (Fig. 2U). Finally, ADSCs were characterized by a higher granularity, as evidenced from cytometric analyses (Fig. 2V, right panel).

Immunophenotyping of MSCs

FC tested ADSCs and DPSCs for the expression of 15 different surface markers. The analyses showed the absence of CD31, CD34, CD45, CD106, and CD117, thus excluding contamination by hematopoietic and endothelial cells (representative FC for negative CD45 is shown in Fig. 3A). Supporting the MSC identity of the cell populations obtained, the frequencies of cells positive for surface markers typical of cultured human MSC, i.e. CD9, CD29, CD44, CD73, CD90 and CD105 exceeded 95% (representative FC for positive CD44 is shown in Fig. 3A). Principal Component Analysis (PCA) based on the median fluorescence intensity (MdfI) values of positive markers demonstrated that tissue origin (adipose tissue vs. dental pulp) was the principal determinant of phenotypic variation, as reflected by the first principal component (PC1, Fig. 3B). On the other hand, within DPSCs, phenotypic heterogeneity was mainly ascribed to the different compartments (coronal vs. radicular). At the same time, the donor source accounted for most of the variability among ADSCs (PC2, Fig. 3B). Hierarchical Cluster Analysis (HCA) confirmed the PCA results (Fig. 3C).

Further analysis showed that ADSCs and DPSCs mainly differ in the expression of five markers: CD10, CD13, CD26, CD56, and CD146 (Fig. 3D). Notably, CD10 and CD26 were either absent or minimally expressed in DPSCs, regardless of the anatomical compartment. In contrast, their expression in ADSCs was donor-dependent: approximately 43% and 90% of cells from one donor expressed CD10 and CD26, respectively, compared to 17–23% and 45–56% in the other donor. Moreover, nearly 100% and only <8% of ADSCs were positive for CD13 and CD56, respectively, whereas DPSCs differed based on tissue compartment. CD13 was expressed in >90% CPSCs and about 31–73% of RPSCs, CD56 in about 49–68% of RPSCs and about 12–34% of CPSCs. Finally, CD146 was expressed in 63–77% of ADSCs and 51–58% of DPSCs.

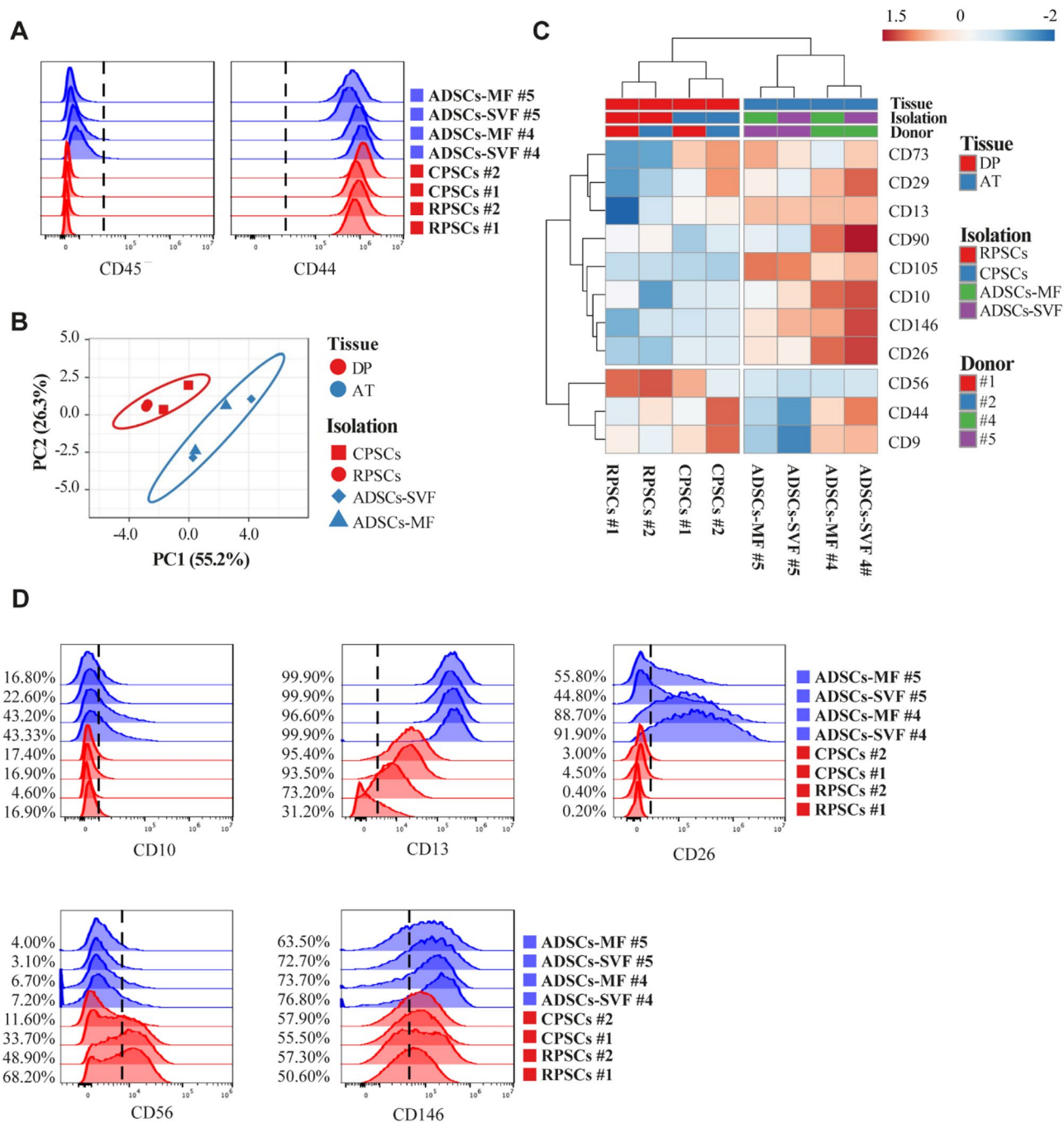


Fig. 3 Flow cytometry analysis of DPSCs and ADSCs. **A**) Histogram overlays showing expression of a negative (CD45) and a positive (CD44) marker in the indicated MSC lines from different donors indicated by numbers (CPSCs #1 and #2, RPSCs #1 and #2, ADSCs-MF #4 and #5, and ADSCs-SVF #4 and #5). Dashed lines indicate boundaries for positivity. **B**) Principal component analysis of the positive markers (CD9, CD10, CD13, CD26, CD29, CD44, CD56, CD73, CD90, CD105, and CD146). After ln-transformation, unit variance scaling was applied to each marker's median fluorescence intensity values; singular value decomposition with imputation was used to calculate principal components (PCs). X and Y axes show PC1 and PC2, explaining 55.2% and 26.3% of the total variance, respectively. Prediction ellipses are traced with probability 0.95 that a new observation from the same group will fall inside the ellipse. **C**) Hierarchical cluster analysis of the surface markers. After ln-transformation, rows were centered and unit variance scaled. Both rows and columns are clustered using Euclidean distance and Ward linkage. **D**) Differentially expressed markers (CD10, CD13, CD26, CD56, and CD146) by the indicated MSC populations

Cytokines, chemokines, and growth factors in the secretome

To analyze the secretory capacity of the ADSCs and DPSCs, cells cultured at 70% confluency were allowed to release their compounds into the media for 3 days (CMs) and 7 days (concentrated culture media, CCMs), the latter following a sequential collection procedure described in Materials and Methods that was developed to verify the possibility of increasing the concentration of the secreted molecules. A multiplexed immunoassay-based platform was used to measure 27 analytes (pro-inflammatory and anti-inflammatory cytokines, chemokines, and growth factors). Detectable levels of all the analytes were found in the CM derived from all cell populations, but with clear diversities among ADSCs and DPSCs. Besides unveiling such differences, the HCA also revealed that whereas the analyte composition of CMs from the dental cell populations (RPSCs and RPSCs) derived from the two donors appeared similar, the isolation method (SVF and MF) was responsible for significant differences among CMs from ADSCs derived from four donors (Fig. 4A). The use of PCA confirmed that the tissue source was the most critical factor of such differences and that the ADSCs differed for the isolation method (Fig. 4B). In line with these observations, statistical analyses

revealed significant differences in the analyte concentration between CMs from DPSCs and ADSCs (Figs. 5 and 6). For instance, eight analytes (IL-4, IL-7, IL-9, IL-13, Eotaxin, MIP-1 β , RANTES, and G-CSF) were expressed at higher amounts in CMs from ADSCs than DPSCs, and conversely, nine analytes (IL-1 β , IL-2, IFN- γ , IL-8, IP-10, MCP1, bFGF, GM-CSF, and VEGF) were higher in CMs from DPSCs than in ADSCs. However, in some of these (IL-1 β , IL-2, IL-8, bFGF), the differences were present only with the CM-SVF and not with the CM-MF of ADSCs, suggesting some dependence on the isolation method, which was mechanical rather than enzymatic. In the same way, whereas no significant differences were observed between CM-RP and CM-CP in DPSCs, the levels of 14 analytes (IL-4, IL-7, IL-9, IL-13, IL-17, TNF- α , Eotaxin, MIP-1 α , MIP-1 β , RANTES, bFGF, G-CSF, GM-CSF, and IL-1ra) were upregulated in the CM-MF with respect to CM-SVF in ADSCs.

Hierarchical Cluster Analysis (Fig. S1 in Additional file 2) performed on the CMs and CCMs showed the persistence of the difference between ADSCs and DPSCs due to the tissue source. Moreover, a trend of factors' accumulation in the CCMs with respect to the relative CMs, both in DPSC and ADSC lines, was evident. Specifically, 16 analytes (IL-1 β , IL-2, IL-5, IL-6, IL-7, IL-8, IL-10, IL-12,

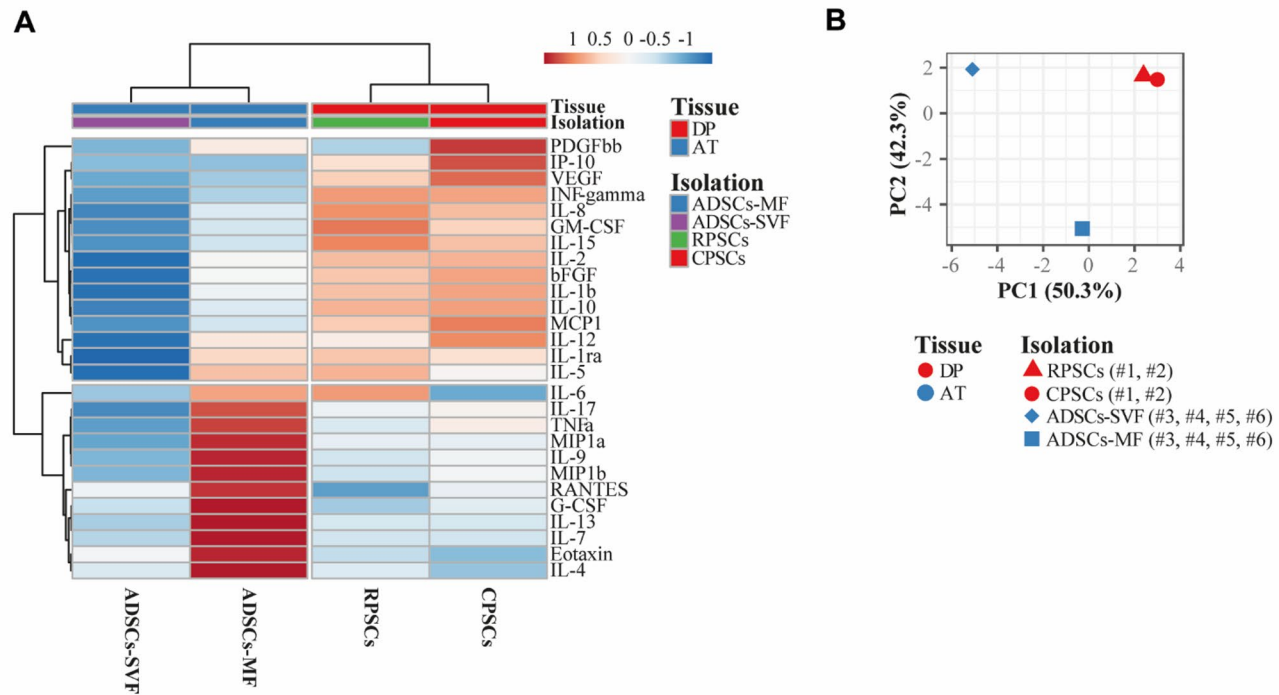


Fig. 4 Cytokines, chemokines, and growth factors secreted in the 3-day CM. **(A)** Hierarchical cluster analysis of cytokines secreted in the CMs by the indicated MSC lines obtained from donors identified by numbers (CPSCs #1 and #2, RPSCs #1 and #2, ADSCs-MF #4 and #5, and ADSCs-SVF #4 and #5). Rows are centered; unit variance scaling is applied to rows. Both rows and columns are clustered using correlation distance and average linkage. 27 rows, 4 columns. **(B)** Principal component analysis. Unit variance scaling is applied to rows; singular value decomposition (SVD) with imputation is used to calculate principal components. X and Y axes show principal component 1 and principal component 2, which explain 50.3% and 42.3% of the total variance, respectively

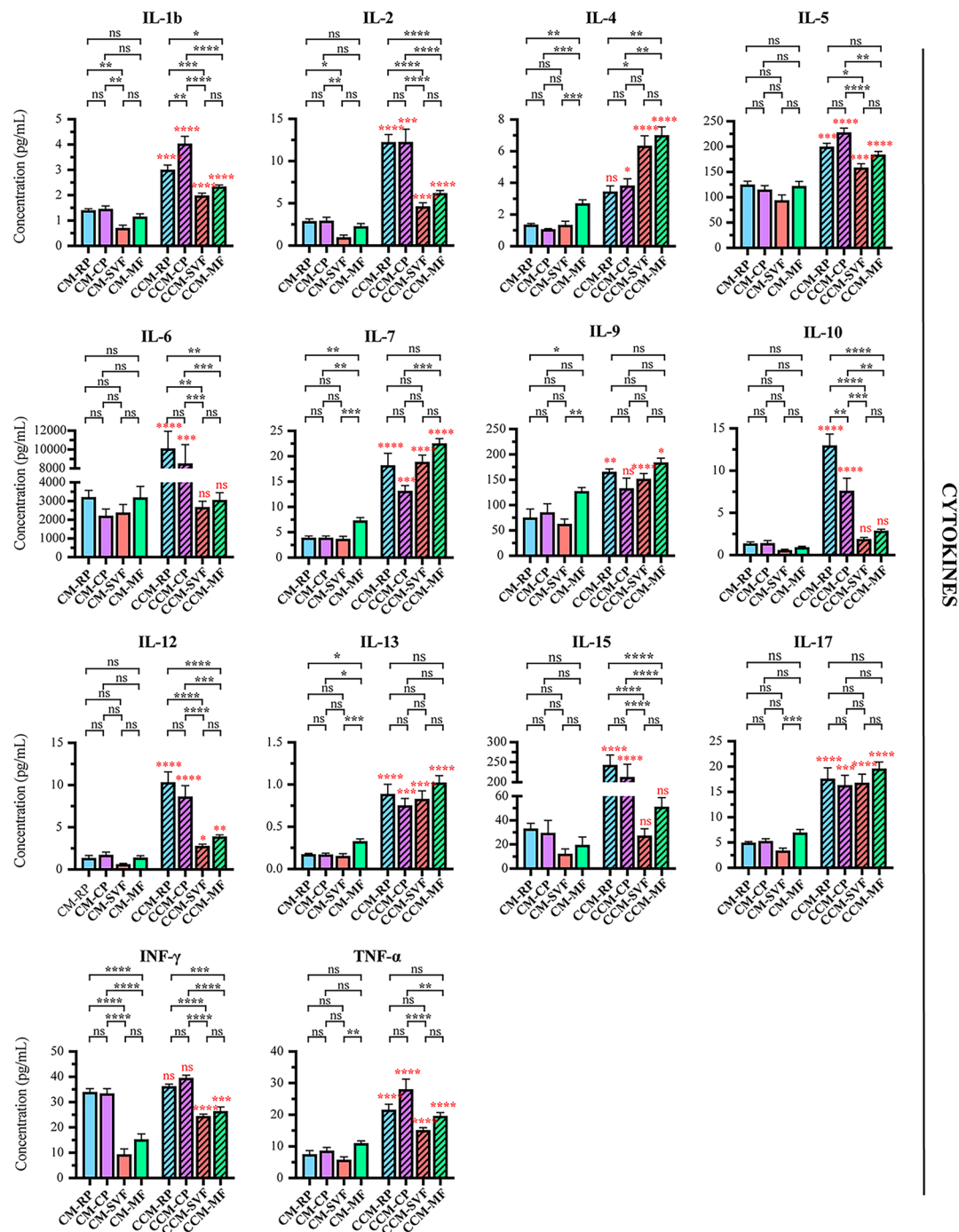


Fig. 5 Bioplex analysis of secreted cytokines in CM and CCM samples of indicated MSCs. Concentration in CM and CCM samples of the cytokines: Interleukin 1 β (IL1 β), Interleukin-2 (IL-2), Interleukin-4 (IL-4), Interleukin-5 (IL-5), Interleukin-6 (IL-6), Interleukin-7 (IL-7), Interleukin-9 (IL-9), Interleukin-10 (IL-10), Interleukin-12 (IL-12), Interleukin-13 (IL-13), Interleukin-15 (IL-15), Interleukin-17 A (IL-17 A), Interferon- γ (IFN- γ) and Tumor necrosis factor- α (TNF- α). Data are expressed as mean $\pm$ SEM. Results of one-way Anova test between the indicated groups (black asterisks) and CM vs. relative CCM (red asterisks) are depicted. ns = not significant, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$

IL-13, IL-15, IL-17, TNF- α , IP-10, IL-1ra, PDGF-bb, and VEGF) were significantly more concentrated in both CCM-RP and CCM-CP with respect to each corresponding CM, whereas three (IL-9, MIP-1 β , GM-CSF) were

significantly upregulated only in the CCM-RP, and three (IL-4, MCP1, bFGF) only in CCM-CP. In CCM obtained from ADSCs, 18 analytes (IL-1 β , IL-2, IL-4, IL-5, IL-7, IL-9, IL-12, IL-13, IL-17, INF γ , TNF- α , Eotaxin, MIP-1 α ,

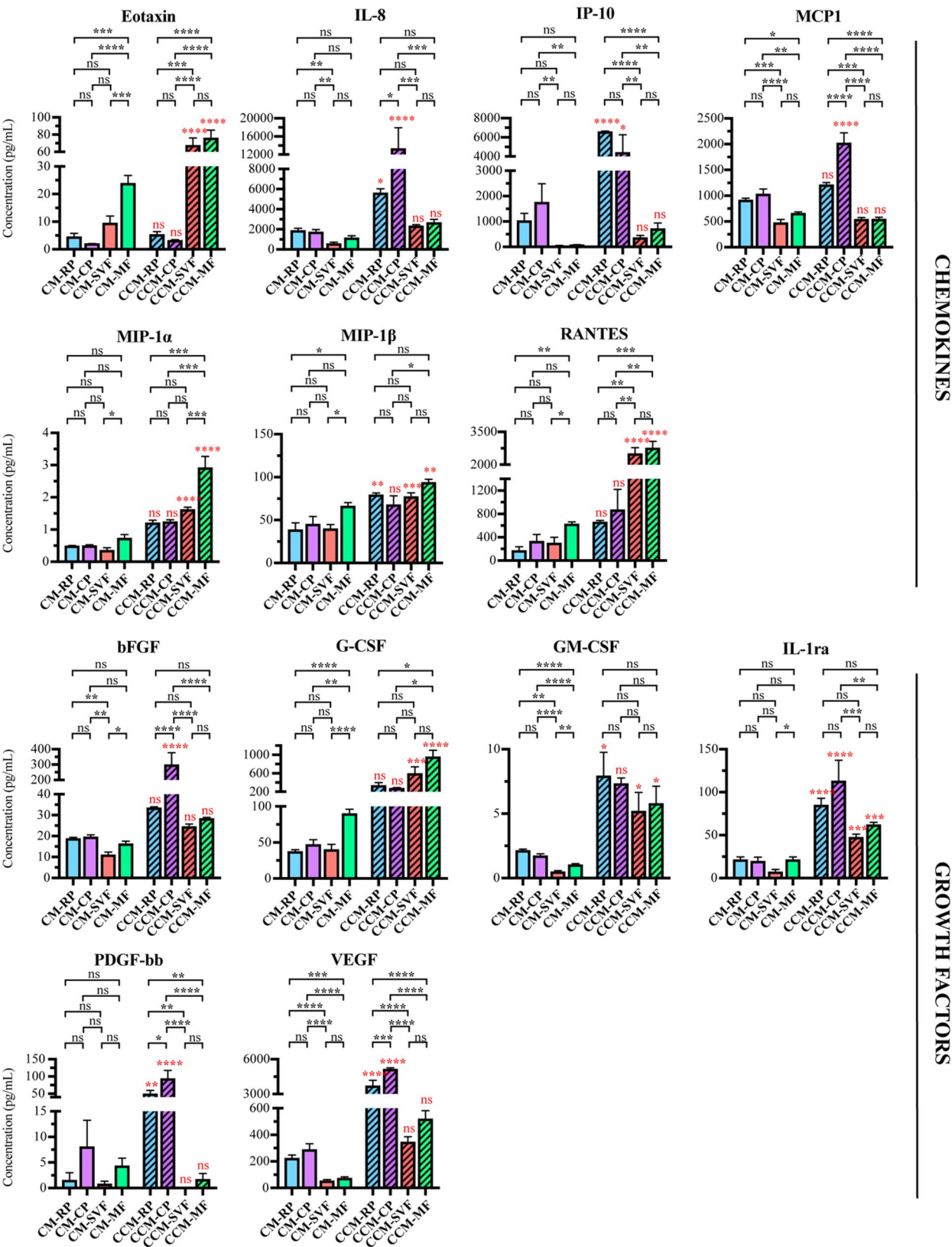


Fig. 6 (See legend on next page.)

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Fig. 6 Bioplex analysis of secreted chemokines and growth factors in CM and CCM samples of MSCs. Concentration in CM and CCM samples of the chemokines: Eotaxin, Interleukin-8 (IL-8), Interferon γ -induced protein 10 (IP-10), Monocyte chemoattractant protein 1 (MCP1), Macrophage inflammatory protein 1 α (MIP-1 α), Macrophage inflammatory protein 1 β (MIP-1 β), Rantes, and the growth factors: basic Fibroblast growth factor (bFGF), Granulocyte colony-stimulating factor (G-CSF), Granulocyte-macrophage colony-stimulating factor (GM-CSF), Interleukin-1 receptor antagonist (IL-1ra), Platelet-derived growth factor bb (PDGF-bb), and Vascular-endothelial growth factor (VEGF). Data are expressed as mean $\pm$ SEM. Results of one-way Anova test between the indicated groups (black asterisks) and CM vs. relative CCM (red asterisks) are depicted. ns = not significant, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$

MIP-1 β , RANTES, G-CSF, GM-CSF, and IL-1ra) were significantly increased in both CCM-SVF and CCM-MF with respect to each corresponding CM (Figs. 5 and 6).

EVs in the secretome

We explored the presence of EVs derived from DPSCs and ADSCs in both CM and CCM. Figure 7 shows the presence of EVs in all the samples analyzed with no detectable difference, in the range of $5\text{--}20 \times 10^6$ EVs/mL (Fig. 7A). Moreover, as expected, dimension analysis (Fig. 7B) confirmed an enrichment of exosome (vesicles < 160 nm in diameter) in all samples of the isolated EVs (CM-RP = $78.66 \pm 1.13\%$; CCM-RP = $68.68 \pm 9.72\%$; CM-CP = $63.53 \pm 0.73\%$; CCM-CP = $71.06 \pm 15.12\%$; CM-SVF = $81.19 \pm 1.48\%$; CCM-SVF = $86.34 \pm 6.57\%$; CM-MF = $78.64 \pm 3.09\%$; CCM-MF = $81.87 \pm 8.22\%$). ADSCs appeared to release a significantly higher number of EVs < 100 nm in diameter compared to DPSCs (CM-SVF = $23.86 \pm 1.85\%$; CCM-SVF = $29.65 \pm 8.02\%$; CM-MF = $23.51 \pm 1.25\%$; CCM-MF = $29.24 \pm 5.75\%$ vs. CM-RP = $10.10 \pm 1.10\%$; CCM-RP = $7.88 \pm 2.23\%$; CM-CP = $5.69 \pm 0.21\%$; CCM-CP = $10.89 \pm 4.49\%$) (Fig. 7B).

microRNAs in the secretome

All cells, particularly MSCs, secrete, besides many proteins, a heterogeneous population of microRNAs (miRNAs). We aimed to identify all miRNAs, free and enclosed in exosomes and microvesicles, present in the secretomes collected from the DPSCs and ADSCs, and compare their expression patterns. We performed this analysis on both CMs (CM-DPSC and CM-ADSC) and CCMs (CCM-DPSC and CCM-ADSC), as done for the secreted analytes evaluated.

After using the protocol for sequencing small RNAs and miRNAs (see Materials and Methods), we selected all high-quality reads with lengths corresponding to mature miRNAs (18–27 nucleotides). This analysis identified and quantified 2,365 mature miRNAs (Raw data in Additional file 3) in both CMs and CCMs. We then investigated whether, and to what extent, the different cell populations, DPSCs and ADSCs, differ in the composition of miRNAs enriching their secretomes.

As illustrated in the Venn diagram (Fig. 8A), evaluating the released miRNAs with at least one read (see Raw data in Additional file 3), the majority (65.4%, 1,455 miRNAs) were shared between CM/CCM-DPSC and

CM/CCM-ADSC. In contrast, 170 miRNAs (7.6%) were uniquely expressed in CM/CCM-DPSC, and 601 miRNAs (27%) were exclusive to CM/CCM-ADSC. These findings suggest that, although the secretomes of DPSCs and ADSCs are broadly similar in miRNA content, CM/CCM-ADSC displays a greater variety of miRNAs than CM/CCM-DPSC. The PCA was performed to project the variance onto the first two principal components, allowing us to simplify the analysis and capture the essential structure of our high-dimensional dataset. The first two components, PC1 and PC2, explained 43% of the total variance (27% and 16%, respectively). Notably, CM/CCM-ADSC samples clustered on the positive side of PC1, while CM/CCM-DPSC samples were located on the negative side, suggesting that the tissue of origin (adipose tissue vs. dental pulp) is the primary factor distinguishing the two populations (Fig. 8B).

The differential expression analysis between CM-ADSC and CM-DPSC based on a fold change > 2.0 ($p < 0.05$) showed that 168 miRNAs displayed significantly different expression (DE) levels with 82 up and 86 down in CM-DPSCs compared with CM-ADSCs, as shown in the Volcano plot (Fig. 8C) and Table S2 (in Additional file 4). To validate this data, we analyzed freshly prepared CMs and measured by RT-qPCR the expression levels of 16 miRNAs, four not DE, and twelve DE, of which six up-regulated in ADSCs and six up-regulated in DPSCs. The results obtained were consistent with omics data, confirming the reliability of the initial screening (Fig. S2 in Additional file 5).

Next, we perform a functional enrichment analysis to identify the main pathways deregulated by the first twenty DE miRNAs between the CM-DPSC and CM-ADSC (Fig. 8D-E). Specifically, we submitted to the Enrichr bioinformatics tool the predicted target genes of the top ten miRNAs upregulated in CM-DPSC compared to CM-ADSC (hsa-miR-483-3p, hsa-miR-483-5p, hsa-miR-191-5p, hsa-miR-1247-5p, hsa-miR-185-5p, hsa-miR-146a-5p, hsa-miR-25-3p, hsa-miR-23a-3p, hsa-miR-197-3p, and hsa-miR-574-3p) (Fig. 8D), as well as those of the top ten miRNAs upregulated in CM-ADSC compared to CM-DPSC (hsa-miR-10a-5p, hsa-miR-143-3p, hsa-miR-10b-5p, hsa-miR-196a-5p, hsa-let-7i-5p, hsa-miR-196b-5p, hsa-miR-615-3p, hsa-miR-224-5p, hsa-miR-10b-3p, and hsa-miR-10a-3p) (Fig. 8E). The enrichment analysis revealed that among the top ten cellular pathways potentially deregulated, six were shared

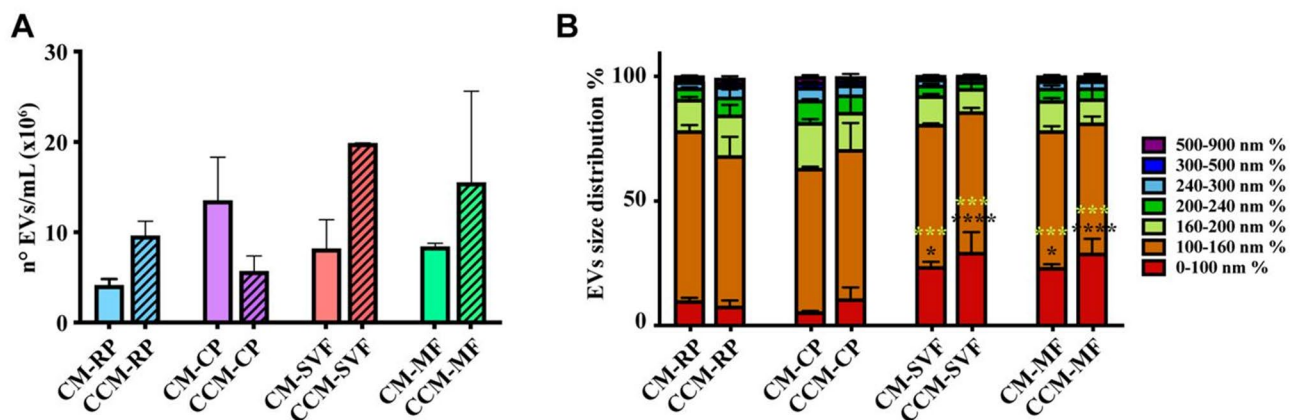


Fig. 7 EVs in the CMs and CCMs of indicated MSCs. **(A)** Concentration of total EVs in CM and CCM samples derived from RPSCs, CPSCs, ADSCs-SVF, and ADSCs-MF. Data are expressed as mean $\pm$ SEM of three independent experiments. **(B)** Size distribution of EVs in the analyzed media. Data are expressed as mean $\pm$ SEM of three independent experiments. Results of one-way Anova test between the SVF/MF-derived media compared with RP media (black asterisks) and CP media (yellow asterisks) are depicted. * $p < 0.05$, *** $p < 0.001$, **** $p < 0.0001$

between ADSCs and DPSCs. These included TNF α signaling, mTORC1 signaling, G2-M checkpoint, protein secretion, UV response (DNA damage), and E2F targets. Four pathways appeared to be specifically affected by DE miRNAs in CM-DPSCs: hypoxia, early estrogen response, p53 signaling, and apoptosis. In contrast, DE miRNAs in CM-ADSCs were associated with the regulation of pathways such as MYC targets V1, mitotic spindle, PI3K/mTOR signaling, and epithelial–mesenchymal transition (EMT).

We also assessed the change in miRNAs between CMs and CCMs with Volcano plots that evidenced a dynamicity in their accumulation and/or degradation (Fig. 8F–G). In particular, 129 miRNAs were differentially expressed between CCM-ADSC and CM-ADSC, of which 56 accumulated after seven days of conditioning, while 73 decreased in their expression (Fig. 8F and Table S3 in Additional file 6). Analogous analysis on DPSCs showed 79 DE miRNAs between CCM-DPSC and CM-DPSC. Among them, 27 miRNAs were enriched and 52 depleted in the CCM (Fig. 8G and Table S4 in Additional file 7).

Discussion

In recent years, the application of MSCs and their derivatives in treating various diseases has received widespread attention. However, there are still multiple problems related to their use. The heterogeneity due to the district of origin and the poor characterization of their secretome represents significant limitations to the clinical applicability.

In the present paper, we obtained eight primary MSC lines from adipose tissue samples of four donors, isolated by enzymatic and mechanical procedures, and four primary cell lines from two donors' dental pulps, each sample divided into coronal and radicular parts, isolated by mechanical methods.

Trilineage differentiation, proliferation assays, and immunophenotyping with cytofluorimetric analysis showed that all cells possessed typical MSC characteristics, although some notable differences exist. While ADSCs could differentiate into osteocytes, chondrocytes, and adipocytes, DPSCs, as reported by others [26, 38], could not produce adipocytes. The reason for this discrepancy emerged in the work of Fracaro and collaborators, in which they demonstrated that, unlike ADSCs, DPSCs showed significantly lower expression levels of the master regulator of adipogenesis, PPAR γ , but substantially higher expression of some WNT pathway genes, such as the wingless-type member 10B (WNT10B) gene that prevented the adipogenic differentiation by blocking the expression of key transcription factors, such as PPAR γ and CEBPA [38]. Since the WNT pathway inhibits adipogenesis but promotes proliferation, this could also explain why DPSCs were significantly more proliferative than ADSCs, displaying a shorter doubling time. It should be considered, however, that the donor's age can also partly explain such findings since the dental pulp donors are younger (18–20-year-olds) than the adipose tissue donors (45–60-year-olds).

In immunofluorimetric analyses, although with some variation among donors, ADSCs presented higher intensities for CD10, CD13, CD26, and CD146. These surface characteristics were linked to a consistently higher doubling time and a higher percentage of α SMA-positive cells than DPSCs. Interestingly, in a study where bone marrow MSCs (BM-MSCs) were evaluated in terms of CD146 expression, the cells showing a higher level of this typical perivascular stem cell marker inside the population of BM-MSCs proliferated significantly slower and were associated with a vascular smooth muscle cell lineage commitment [39], showing pericyte-like progenitor characteristics [40]. Overall, these results, along with the

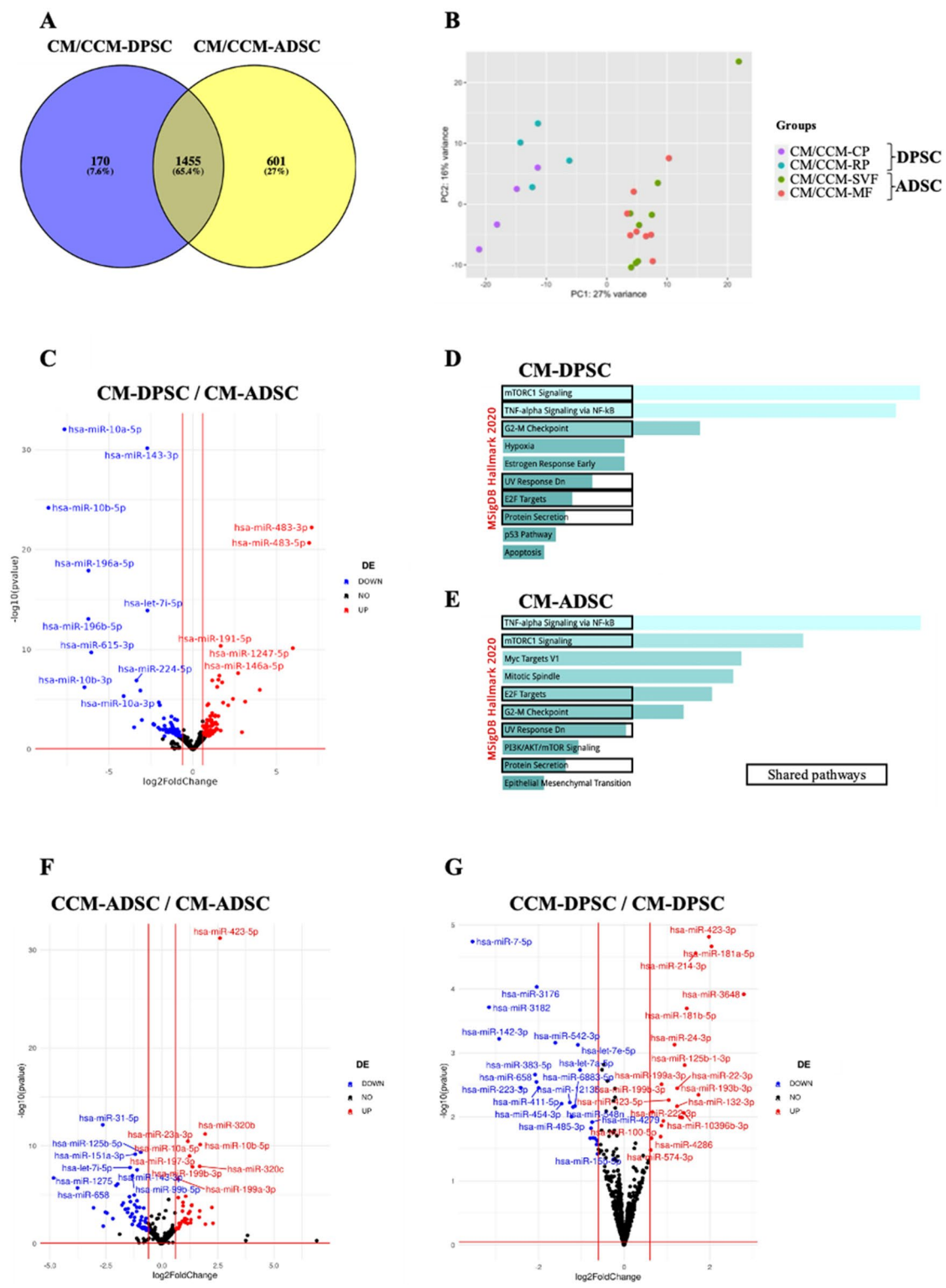


Fig. 8 (See legend on next page.)

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Fig. 8 Analysis of miRNAs expressed in CMs and CCMs (CM/CCM). **(A)** A Venn diagram showing the overlap of miRNAs with at least one read (see Materials and Methods) in the miRNome secreted in CM/CCM obtained by all DPSCs and ADSC populations. **(B)** Principal component analysis of miRNAs enriched in the CMs after conditioning them for 3 days (CM-CP, CM-RP, CM-SVF, CM-MF) and 7 days (CCM-CP, CCM-RP, CCM-SVF, CCM-MF) obtained by all cell populations indicated by dots. The statistical units are projected in the space spanned by the two major components. **(C)** Volcano plot of differentially expressed miRNAs between CM-DPSC and CM-ADSC. **(D)** Molecular signatures database (MSigDB) functional annotation of the Top ten pathways deregulated by miRNAs secreted in DPSC-CM. **(E)** Molecular signatures database (MSigDB) functional annotation of the Top ten pathways deregulated by miRNAs secreted in ADSC-CM. **(F)** Volcano plot of differentially expressed miRNAs between CCM-ADSC and CM-ADSC. **(G)** Volcano plot of differentially expressed miRNAs between CCM-DPSC and CM-DPSC. Each volcano plot shows the log2 fold change and $p < 0,05$ as threshold for significance. Red and blue dots indicate upregulated and downregulated differentially expressed mRNAs, respectively

PCA and HCA algorithms performed on cytofluorimetric data, suggest that ADSC lines possess more typical stem/progenitor MSC characteristics in comparison to DPSCs and that donor diversity (also for gender) rather than isolation methods can be a significant source of CD-protein ADSC heterogeneity. In contrast, DPSCs showed limited CD-protein heterogeneity due to their regional (coronal or radicular) origin inside the dental pulp. Surprisingly, the radicular cell populations derived from the two donors (both females) are almost superimposable in the PCA analysis. As evidenced by HCA, these cells present a lower level of nearly all surface markers, except for CD56. This membrane protein, known as Neural-Cell Adhesion Molecule, is involved in cell-cell interactions and neural stem cell migration during development and differentiation along the neural lineage [41], confirming the neuro-ectodermal origin of these MSCs, which are probably earlier progenitors in comparison with their coronal counterpart. In addition, the fact that >90% of DPSCs were Nestin positive compared to only about 1% of ADSCs suggests stemness differences. Nestin expression in nearly all DPSCs, already reported in the literature [42], indicates that these cells might be more likely to differentiate into cell types of various embryological origins other than mesodermal, like neuronal and endothelial cells [43], in line with their neuro-ectodermal origin from the neural crest [44].

As far as we know, this is the first time that the separation between radicular and coronal parts of dental pulp has been made, starting from open apex third molars that were deliberately chosen to assess whether a more proliferating and less differentiated population was present in the radicular region close to the apex that was not yet fully formed. Based on the proliferation data, this seems to be the case. However, there are no differences between the two dental populations, coronal and radicular, in terms of differentiation capacity, at least as far as the mesodermal lineage is concerned. It would be interesting, in the future, to explore other lineage differentiation capabilities, such as neuronal ones, since the presence of Nestin makes neuroectodermal derivation evident in both dental cell populations. Moreover, we would have liked to evaluate the two types of extraction methodologies in the dental pulp. Still, given the limited material in the tooth, further division was not possible after

an initial regional division (coronal and apical). Since we have found in the literature that an evaluation of the two different extraction methods had already been conducted on dental pulp, albeit not as extensively as our analysis, and had not indicated significant differences between the two populations obtained [32], this was the reason why we focused, in third molars with incompletely formed apices, on the possible presence of a difference due to regional location, therefore dividing the coronal portion from the apical one, never done before. Thus, this represents both a novelty and a limitation of the study.

It is known that adult MSCs exhibit various paracrine functions and, depending on tissue sources, differ significantly in their immunomodulatory, anti-inflammatory, and regenerative capabilities [45]. Given their importance for prospective clinical applications, we were primarily interested in the secretomes produced by these cell lines in culture. It is known that it consists of soluble molecules as growth factors (i.e., VEGF, TGF- β , HGF, FGF), chemokines (i.e., CXCL12, CCL2), and immunomodulatory cytokines (i.e., IL-10, IDO, TSG6, PGE2) [46], in addition to microRNA and microvesicles [47], that confer pro-angiogenic, anti-inflammatory, and immune suppressive activities onto a variety of cells and tissues [48].

Using a multiplex immunoassay for the measurement of pro-inflammatory and anti-inflammatory cytokines, chemokines, and growth factors, we found that 3-day CM from all cell lines contained detectable levels of the tested analytes, but with specific concentration differences among ADSCs and DPSCs. Interestingly, PCA and HCA algorithms showed that the secretomes from ADSCs mainly differed based on isolation methods rather than donors, unlike in surface marker analysis. This is quite an interesting point. In the adipose tissue, where the two extraction methods have been used on the same tissue samples and comparison can be made, MSCs obtained by MF secrete cytokines, chemokines, and growth factors to a greater extent than their counterparts obtained by enzymatic digestion. This might be related to how cells are recovered during the MF procedure, which comprises an explant culture, in which cells grow out from the piece of tissue since they are in some way “activated” in the stem cell niche, where the ECM behaves as in a wound healing [49, 50]. These differences persisted after seven days of culture, showing various degrees of

analyte accumulation. On the other hand, no significant differences in the secreted factors were detected between media from CPSCs and RPSCs. Since the two populations were obtained by the MF method, this supports the hypothesis that different extraction procedures could have influenced the quantity of secreted molecules, as we have found in adipose samples, at least for those analytes we evaluated in this study, which are related to immunomodulation and inflammation.

Based on the overall analysis, the presence of a consistent amount of pro-inflammatory and anti-inflammatory cytokines and chemokines makes it possible to predict that all CMs, regardless of the source, can modulate immune responses. In addition, the occurrence of GM-CSF, IP10 (CXCL10), and INF- γ , as well as of VEGF and bFGF, mostly in CMs obtained from DPSCs, suggests a more consistent immunomodulatory as well as regenerative (angiogenesis/wound healing) effect of these media [51].

The analysis of miRNA in the CM/CCM showed that organ source (dental pulp vs. adipose tissue) was the most critical diverging factor among the miRNomes analyzed. However, it should be considered that we preferred to evaluate the miRNA content of the entire CM, not just that present in the extravesicular compartment, since it is known that miRNAs can “travel” between cells even outside of EVs/exosomes [52]. Furthermore, the observed accumulation and/or degradation of miRNAs after seven days of medium conditioning underscores the dynamic nature of their secretion and/or differences in their stability.

On the collected data, we utilized the public database TarBase v9.0 to identify target genes of the top twenty differentially expressed miRNAs between CM-DPSC and CM-ADSC, evenly split with ten upregulated in each. These target genes, analyzed using EnrichR, revealed that pathways commonly deregulated by both secretomes are primarily involved in inflammatory processes, protein translation, cell cycle checkpoints, E2F target regulation of the cell cycle and DNA synthesis, and DNA repair following UV-induced damage. Moreover, signaling pathways uniquely dysregulated by CM-DPSC are predominantly involved in oxidative stress, apoptosis, and early estrogen response. This is consistent with the enrichment of specific miRNAs such as hsa-miR-143-3p, previously reported to induce granulosa cell apoptosis by targeting BMPRI1A in polycystic ovary syndrome [53].

Similarly, hsa-miR-196a-5p has been implicated in facilitating the progression of estrogen-dependent endometrial carcinoma via FOXO1 regulation [54], while hsa-miR-615-3p has demonstrated protective effects against oxidative stress-induced cardiomyocyte injury through MEF2A modulation [55]. These findings suggest that CM-DPSC secretomes may positively affect cellular

responses to oxidative damage and hormone-mediated signaling.

In contrast, the CM-ADSC secretome appeared more closely associated with pathways regulating cell proliferation, cell cycle control, and epithelial–mesenchymal transition (EMT). Notably, miRNAs such as hsa-miR-146a-5p, known to promote EMT in esophageal squamous cell carcinoma via NOTCH-2 targeting [56], and hsa-miR-483-3p, which inhibits cell cycle progression by targeting CDC25A and reducing CCND–CDK4/6 complex assembly [57], were highly expressed in CM-ADSC. These observations support the notion that CM-ADSC-derived factors may contribute more substantially to regulating proliferative and differentiation processes. These data highlight distinct functional specializations of ADSC and DPSC secretomes, reflecting their tissue origins and potentially informing their respective therapeutic applications. Meaning and functional/applicative consequence of the difference between ADSCs and DPSCs reported above is to be investigated further, opening new scenarios for future interventions.

Finally, since human MSC-derived EVs have shown their therapeutic effect in preclinical models [58], we have investigated their presence in the CM of the produced cell lines using tools that measure their concentration in samples and molecular composition. We found that all CM/CCMs contained EVs with no significant differences in concentration among samples. However, the analysis revealed that ADSCs appeared to release a significantly higher percentage of small exosomes (< 100 nm in diameter) than DPSCs. We can hypothesize that the higher number of smaller exosomes could determine the higher diversity in miRNA species we have found in CM/CCM from ADSCs, as indicated by the Venn diagram. We do not know whether and to what extent this diversity might impact the therapeutic potential, and this issue will be interesting to investigate further.

Conclusions

This study presents an in-depth analysis and comparison of human MSCs obtained from two adult connective tissues with different ontogenies. Together, these data confirm the heterogeneity of MSCs based on anatomical location and isolation methods, apart from age and gender, and the possibility to produce secretomes rich in cytokines, chemokines, growth factors, EVs, and microRNAs from these cells. Despite the limitations due to the number of human tissue samples analyzed, these results represent the basis for improving and deepening MSC characterization, selection, and controlled production of CMs, which are more appropriate for specific therapeutic targets to enhance MSC-based therapies consistently and reliably.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13062-025-00697-v>.

Supplementary Material 1: Table S1. Flow cytometry reagents. Information on antibodies utilized in the cytofluorimetric analysis (isotype, fluorochrome, company, dilution, code number).

Supplementary Material 2: Fig. S1. Cytokines, chemokines, and growth factors expression in the concentrated conditioned medium (CCM). Hierarchical cluster analysis of cytokines expressed by the indicated cell populations in the CCMs. Rows are centered; unit variance scaling is applied to rows. Rows are clustered using correlation distance and average linkage. Columns are clustered using correlation distance and single linkage. Twenty-seven rows, eight columns.

Supplementary Material 3: Raw data from the miRNome analysis for the CMs and CCMs of the 12 primary cell lines under study

Supplementary Material 4: Table S2. Differentially expressed miRNAs between CM-ADSC and CM-DPSC and relative expression counts with pVal adj.

Supplementary Material 5: Fig. S2. miRNome validation Description: RT-qPCR was performed on sixteen miRNAs (six upregulated in DPSC-CM, six upregulated in ADSC-CM, and four not differentially expressed between the two conditions). Results of multiple unpaired t-test between DPSC-CM and ADSC-CM (*p < 0.05, **p < 0.01, ***p < 0.001, and ****p < 0.0001).

Supplementary Material 6: Table S3. Differentially expressed miRNAs between CM-ADSC and CCM-ADSC and relative expression counts with pVal adj.

Supplementary Material 7: Table S4. Differentially expressed miRNAs between CM-DPSC and CCM-DPSC and relative expression counts with pVal adj.

Acknowledgements

We thank Dr Cosimo Tudisco for providing discarded human samples and Gabriele Rossi for preparing and staining histological sections.

Author contributions

AV, SM, AL, and MAU contributed to the experimental design, data analysis, and interpretation; AV, SM, and MAU contributed to the collection and assembly of data and manuscript drafting; GS, AG, SP, MM, MP, RR, and SV contributed to the collection and assembly of data and data analysis; DF, ACh, GB, MDF, FGK, and VC contributed to the data interpretation and manuscript revision; AC: Conception and design, Data analysis and interpretation, Manuscript writing and revision. All authors read and approved the final manuscript.

Funding

The Italian Ministry of University and Research supported the study through PRIN grants N. 20209L8BN4 and N. 2022A24YYY.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Human abdominal adipose tissue biopsies were harvested at the Department of Orthopedics at the Policlinico Tor Vergata, according to the policies approved by the Ethical Committee of Fondazione PTV Policlinico Tor Vergata with authorization number 160/20. Human third molar teeth were harvested at the Dental Clinic of Policlinico Tor Vergata, according to the policies approved by the Ethical Committee of Fondazione PTV Policlinico Tor Vergata with authorization number 172/22.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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Received: 11 August 2025 / Accepted: 6 October 2025

Published online: 13 November 2025

References

1. Ullah I, Subbarao RB, Rho GJ. Human mesenchymal stem cells - current trends and future prospective. *Biosci Rep* [Internet]. 2015 Apr 28 [cited 2025 Mar 25];35(2):e00191. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4413017/>
2. Dominici M, Le Blanc K, Mueller I, Slaper-Cortenbach I, Marini FC, Krause DS, et al. Minimal criteria for defining multipotent mesenchymal stromal cells. The international society for cellular therapy position statement. *Cytotherapy* [Internet]. 2006 Jan 1 [cited 2025 Mar 25];8(4):315–7. Available from: <https://www.sciencedirect.com/science/article/pii/S1465324906708817>
3. Viswanathan S, Shi Y, Galipeau J, Krampere M, Leblanc K, Martin I, et al. Mesenchymal stem versus stromal cells: international society for cell & gene therapy (ISCT®) mesenchymal stromal cell committee position statement on nomenclature. *Cytotherapy*. 2019;21(10):1019–24.
4. Vizoso FJ, Eiro N, Cid S, Schneider J, Perez-Fernandez R. Mesenchymal stem cell secretome: toward cell-free therapeutic strategies in regenerative medicine. *Int J Mol Sci*. 2017;18(9):1852.
5. Sagaradze G, Grigorieva O, Nimiritsky P, Basalova N, Kalinina N, Akopyan Z, et al. Conditioned medium from human mesenchymal stromal cells: towards the clinical translation. *Int J Mol Sci*. 2019;20(7):1656.
6. Kandoi LPK, Misra S, Verma RSVK. The mesenchymal stem cell secretome: A new paradigm towards cell-free therapeutic mode in regenerative medicine. *Cytokine Growth Factor Rev*. 2019;46:1–9.
7. Nguyen-Truong M, Hematti P, Wang Z. Current status of myocardial restoration via the paracrine function of mesenchymal stromal cells. *Am J Physiol Heart Circ Physiol*. 2021 July 1;321(1):H112–27.
8. Liu A, Zhang X, He H, Zhou L, Naito Y, Sugita S, et al. Therapeutic potential of mesenchymal stem/stromal cell-derived secretome and vesicles for lung injury and disease. *Expert Opin Biol Ther*. 2020;20(2):125–40.
9. Chen YT, Tsai MJ, Hsieh N, Lo MJ, Lee MJ, Cheng H, et al. The superiority of conditioned medium derived from rapidly expanded mesenchymal stem cells for neural repair. *Stem Cell Res Ther*. 2019;10(1):390.
10. Jabbehadri S, Yazdanpanah G, Kanu LN, Chen E, Kang K, Anwar KN, et al. Therapeutic effects of lyophilized Conditioned-Medium derived from corneal mesenchymal stromal cells on corneal epithelial wound healing. *Curr Eye Res*. 2020;45(12):1490–6.
11. Veronesi F, Borsari V, Sartori M, Orciani M, Mattioli-Belmonte M, Fini M. The use of cell conditioned medium for musculoskeletal tissue regeneration. *J Cell Physiol*. 2018 June;233(6):4423–42.
12. Harrell CR, Fellabaum C, Jovicic N, Djonov V, Arsenijevic N, Volarevic V. Molecular mechanisms responsible for therapeutic potential of mesenchymal stem Cell-Derived secretome. *Cells*. 2019;8(5):467.
13. Castro-Manrreza ME, Montesinos JJ. Immunoregulation by mesenchymal stem cells: biological aspects and clinical applications. *J Immunol Res*. 2015;2015:394917.
14. Gao F, Chiu SM, Motan DaL, Zhang Z, Chen L, Ji HL, et al. Mesenchymal stem cells and immunomodulation: current status and future prospects. *Cell Death Dis*. 2016;7(1):e2062.

15. Kupcova Skalninkova H. Proteomic techniques for characterisation of mesenchymal stem cell secretome. *Biochimie*. 2013;95(12):2196–211.
16. Phinney DG, Pittenger MF. Concise review: MSC-Derived exosomes for Cell-Free therapy. *Stem Cells*. 2017;35(4):851–8.
17. Ferguson SW, Wang J, Lee CJ, Liu M, Neelamegham S, Canty JM, et al. The MicroRNA regulatory landscape of MSC-derived exosomes: a systems view. *Sci Rep*. 2018;8(1):1419.
18. Dostert G, Mesure B, Menu P, Velot É. How do mesenchymal stem cells influence or are influenced by microenvironment through extracellular vesicles communication? *Front Cell Dev Biol*. 2017;5:6.
19. Lai RC, Yeo RWY, Lim SK. Mesenchymal stem cell exosomes. *Semin Cell Dev Biol*. 2015;40:82–8.
20. Shin S, Lee J, Kwon Y, Park KS, Jeong JH, Choi SJ, et al. Comparative proteomic analysis of the mesenchymal stem cells secretome from Adipose, bone Marrow, placenta and wharton's jelly. *Int J Mol Sci*. 2021;22(2):845.
21. Wang Zgang, He Z yi, Liang S, Yang Q, Cheng P, Chen A min. Comprehensive proteomic analysis of exosomes derived from human bone marrow, adipose tissue, and umbilical cord mesenchymal stem cells. *Stem Cell Research & Therapy* [Internet]. 2020 Nov 27 [cited 2025 Mar 25];11(1):511. Available from: <https://doi.org/10.1186/s13287-020-02032-8>
22. Tachida Y, Sakurai H, Okutsu J, Suda K, Sugita R, Yaginuma Y, et al. Proteomic comparison of the secreted factors of mesenchymal stem cells from bone marrow, adipose tissue and dental pulp. *J Proteomics Bioinform* [Internet]. [cited 2025 Mar 25];8(12):1–8. Available from: <https://www.longdom.org/>
23. Al-Ghadban S, Artiles M, Bunnell BA. Adipose stem cells in regenerative medicine: looking forward. *Front Bioeng Biotechnol*. 2021;9:837464.
24. Gentile P, Orlandi A, Sciolli MG, Di Pasquali C, Bocchini I, Cervelli V. Concise review: adipose-derived stromal vascular fraction cells and platelet-rich plasma: basic and clinical implications for tissue engineering therapies in regenerative surgery. *Stem Cells Transl Med* [Internet]. 2012 Mar [cited 2025 Mar 25];1(3):230–6. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3659840/>
25. Mohamed-Ahmed S, Fristad I, Lie SA, Suliman S, Mustafa K, Vindenes H, et al. Adipose-derived and bone marrow mesenchymal stem cells: a donor-matched comparison. *Stem Cell Res Ther* [Internet]. 2018 June 19 [cited 2025 Mar 25];9(1):168. Available from: <https://doi.org/10.1186/s13287-018-0914-1>
26. Monterubbianesi R, Bencun M, Pagella P, Woloszyk A, Orsini G, Mitsiadis TA. A comparative in vitro study of the osteogenic and adipogenic potential of human dental pulp stem cells, gingival fibroblasts and foreskin fibroblasts. *Sci Rep* [Internet]. 2019 Feb 11 [cited 2025 Mar 25];9(1):1761. Available from: <http://www.nature.com/articles/s41598-018-37981-x>
27. Yamada Y, Nakamura-Yamada S, Kusano K, Baba S. Clinical potential and current progress of dental pulp stem cells for various systemic diseases in regenerative medicine: A concise review. *Int J Mol Sci*. 2019;20(5):1132.
28. Gancheva MR, Kremer KL, Gronthos S, Koblar SA. Using dental pulp stem cells for stroke therapy. *Front Neurol* [Internet]. 2019 Apr 29 [cited 2025 Mar 25];10:422. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6501465/>
29. Gronthos S, Mankani M, Brahmi J, Robey PG, Shi S. Postnatal human dental pulp stem cells (DPSCs) in vitro and in vivo. *Proc Natl Acad Sci U S A*. 2000;97(25):13625–30.
30. Tremolada C, Colombo V, Ventura C. Adipose tissue and mesenchymal stem cells: state of the Art and Lipogems® technology development. *Curr Stem Cell Rep*. 2016;2(3):304–12.
31. Bianchi F, Maioli M, Leonardi E, Olivi E, Pasquinelli G, Valente S, et al. A new nonenzymatic method and device to obtain a fat tissue derivative highly enriched in pericyte-like elements by mild mechanical forces from human lipoaspirates. *Cell Transpl*. 2013;22(11):2063–77.
32. Hilkens P, Gervois P, Fanton Y, Vanormelingen J, Martens W, Struys T, et al. Effect of isolation methodology on stem cell properties and multilineage differentiation potential of human dental pulp stem cells. *Cell Tissue Res* [Internet]. 2013 July [cited 2025 Aug 7];353(1):65–78. Available from: <http://link.springer.com/https://doi.org/10.1007/s00441-013-1630-x>
33. Salvatore G, De Felici M, Dolci S, Tudisco C, Cicconi R, Campagnolo L, et al. Human adipose-derived stromal cells transplantation prolongs reproductive lifespan on mouse models of mild and severe premature ovarian insufficiency. *Stem Cell Res Ther* [Internet]. 2021 Oct 10 [cited 2025 Aug 7];12(1):537. Available from: <https://stemcellres.biomedcentral.com/articles/https://doi.org/10.1186/s13287-021-02590-5>
34. Caggiati A, Germani A, Di Carlo A, Borsellino G, Capogrossi MC, Picozza M. Naturally adipose stromal Cell-Enriched fat graft: comparative polychromatic flow cytometry study of fat harvested by barbed or blunt multihole cannula. *Aesthet Surg J*. 2017;37(5):591–602.
35. Metsalu T, Vilo J. ClustVis: a web tool for visualizing clustering of multivariate data using principal component analysis and heatmap. *Nucleic Acids Res*. 2015 July 1;43(W1):W566–570.
36. Russo R, Vassallo V, Stellavato A, Valletta M, Cimini D, Pedone PV, et al. Differential secretome profiling of human osteoarthritic synoviocytes treated with biotechnological unsulfated and marine sulfated chondroitins. *Int J Mol Sci* [Internet]. 2020 May 26 [cited 2025 Aug 7];21(11):3746. Available from: <https://www.mdpi.com/1422-0067/21/11/3746>
37. Vumbaca S, Giuliani G, Fiorentini V, Tortolici F, Cerquone Perpetuini A, Riccio F, et al. Characterization of the skeletal muscle secretome reveals a role for extracellular vesicles and IL1 α /IL1 β in restricting fibro/adipogenic progenitor adipogenesis. *Biomolecules* [Internet]. 2021 Aug 8 [cited 2025 Mar 25];11(8):1171. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8392554/>
38. Fracaro L, Senegaglia AC, Herai RH, Leitolis A, Boldrini-Leite LM, Rebelatto CLK, et al. The expression profile of dental Pulp-Derived stromal cells supports their limited capacity to differentiate into adipogenic cells. *Int J Mol Sci*. 2020;21(8):2753.
39. Espagnolle N, Guilloton F, Deschaseaux F, Gadelorge M, Sensébé L, Bourin P. CD 146 expression on mesenchymal stem cells is associated with their vascular smooth muscle commitment. *J Cell Mol Med* [Internet]. 2014 Jan [cited 2025 Aug 8];18(1):104–14. Available from: <https://doi.org/10.1111/jcm.12168> <https://onlinelibrary.wiley.com/doi/>
40. Wang Y, Xu J, Chang L, Meyers CA, Zhang L, Broderick K et al. Relative contributions of adipose-resident CD146+ pericytes and CD34+ adventitial progenitor cells in bone tissue engineering. *npj Regen Med* [Internet]. 2019 Jan 7 [cited 2025 Aug 8];4(1):1. Available from: <https://www.nature.com/articles/s41536-018-0063-2>
41. Hombach-Klonisch S, Panigrahi S, Rashedi I, Seifert A, Alberti E, Pocar P, et al. Adult stem cells and their trans-differentiation potential—perspectives and therapeutic applications. *J Mol Med* [Internet]. 2008 Dec [cited 2025 Aug 8];86(12):1301–14. Available from: <http://link.springer.com/https://doi.org/10.1007/s00109-008-0383-6>
42. Bernal A, Arranz L. Nestin-expressing progenitor cells: function, identity and therapeutic implications. *Cell Mol Life Sci* [Internet]. 2018 June [cited 2025 Aug 8];75(12):2177–95. Available from: <http://link.springer.com/https://doi.org/10.1007/s00018-018-2794-z>
43. Mattei V, Martellucci S, Pulcini F, Santilli F, Sorice M, Delle Monache S. Regenerative Potential of DPSCs and Revascularization: direct, paracrine or autocrine effect? *Stem Cell Rev and Rep* [Internet]. 2021 Oct [cited 2025 Aug 8];17(5):1635–46. Available from: <https://link.springer.com/https://doi.org/10.1007/s12015-021-10162-6>
44. Solis-Castro OO, Rivolta MN, Boissonade FM. Neural Crest-Derived Stem Cells (NCSCs) Obtained from Dental-Related Stem Cells (DRSCs): A literature review on current knowledge and directions toward translational applications. *IJMS* [Internet]. 2022 Feb 28 [cited 2025 Aug 8];23(5):2714. Available from: <https://www.mdpi.com/1422-0067/23/5/2714>
45. Li J, Wu Z, Zhao L, Liu Y, Su Y, Gong X, et al. The heterogeneity of mesenchymal stem cells: an important issue to be addressed in cell therapy. *Stem Cell Res Ther* [Internet]. 2023 Dec 20 [cited 2025 Aug 8];14(1):381. Available from: <https://stemcellres.biomedcentral.com/articles/https://doi.org/10.1186/s13287-023-03587-y>
46. Li P, Ou Q, Shi S, Shao C. Immunomodulatory properties of mesenchymal stem cells/dental stem cells and their therapeutic applications. *Cell Mol Immunol* [Internet]. 2023 Mar 27 [cited 2025 Aug 8];20(6):558–69. Available from: <https://www.nature.com/articles/s41423-023-00998-y>
47. Asgarpour K, Shojaei Z, Amir F, Ai J, Mahjoubin-Tehrani M, Ghasemi F et al. Exosomal microRNAs derived from mesenchymal stem cells: cell-to-cell messages. *Cell Commun Signal* [Internet]. 2020 Dec [cited 2025 Aug 8];18(1):149. Available from: <https://biosignaling.biomedcentral.com/articles/https://doi.org/10.1186/s12964-020-00650-6>
48. Kehl D, Generali M, Mallone A, Heller M, Uldry AC, Cheng P, et al. Proteomic analysis of human mesenchymal stromal cell secretomes: a systematic comparison of the angiogenic potential. *npj Regen Med* [Internet]. 2019 Apr 16 [cited 2025 Apr 28];4(1):8. Available from: <https://www.nature.com/articles/s41536-019-0070-y>
49. Schultz GS, Wysocki A. Interactions between extracellular matrix and growth factors in wound healing. *Wound Repair Regeneration* [Internet]. 2009 Mar [cited 2025 Aug 8];17(2):153–62. Available from: <https://onlinelibrary.wiley.com/doi/https://doi.org/10.1111/j.1524-475X.2009.00466.x>

50. Hendijani F. Explant culture: An advantageous method for isolation of mesenchymal stem cells from human tissues. *Cell Proliferation* [Internet]. 2017 Apr [cited 2025 Aug 8];50(2):e12334. Available from: <https://onlinelibrary.wiley.com/doi/https://doi.org/10.1111/cpr.12334>
51. Li F, Wang X, Shi J, Wu S, Xing W, He Y. Anti-inflammatory effect of dental pulp stem cells. *Front Immunol* [Internet]. 2023 Nov 23 [cited 2025 Aug 8];14:1284868. Available from: <https://www.frontiersin.org/articles/https://doi.org/10.3389/fimmu.2023.1284868/full>
52. Vickers KC, Palmisano BT, Shoucri BM, Shamburek RD, Remaley AT. MicroRNAs are transported in plasma and delivered to recipient cells by high-density lipoproteins. *Nat Cell Biol* [Internet]. 2011 Apr [cited 2025 Aug 8];13(4):423–33. Available from: <https://www.nature.com/articles/ncb2210>
53. Zhao Y, Pan S, Li Y, Wu X. Exosomal miR-143-3p derived from follicular fluid promotes granulosa cell apoptosis by targeting BMPR1A in polycystic ovary syndrome. *Sci Rep* [Internet]. 2022 Mar 14 [cited 2025 Aug 8];12(1):4359. Available from: <https://www.nature.com/articles/s41598-022-08423-6>
54. Zhu Y, Tang Y, Fan Y, Wu D. MiR-196a-5p facilitates progression of estrogen-dependent endometrial cancer by regulating FOXO1. *Histol Histopathol*. 2023;38(10):1157–68.
55. Zhang D, Zhang G, Yu K, Zhang X, Jiang A. MiRNA-615-3p alleviates oxidative stress injury of human cardiomyocytes via PI3K/Akt signaling by targeting MEF2A. *Anatol J Cardiol*. 2022;26(5):373–81.
56. Wang C, Zhang W, Zhang L, Chen X, Liu F, Zhang J, et al. miR-146a-5p mediates epithelial–mesenchymal transition of oesophageal squamous cell carcinoma via targeting Notch2. *Br J Cancer* [Internet]. 2016 Dec 6 [cited 2025 Aug 8];115(12):1548–54. Available from: <https://www.nature.com/article/s/bjc2016367>
57. Bertero T, Gastaldi C, Bourget-Ponzio I, Mari B, Meneguzzi G, Barbry P, et al. CDC25A targeting by miR-483-3p decreases CCND–CDK4/6 assembly and contributes to cell cycle arrest. *Cell Death Differ* [Internet]. 2013 June [cited 2025 Aug 8];20(6):800–11. Available from: <https://www.nature.com/articles/cdd20135>
58. Kou M, Huang L, Yang J, Chiang Z, Chen S, Liu J, et al. Mesenchymal stem cell-derived extracellular vesicles for immunomodulation and regeneration: a next generation therapeutic tool? *Cell Death Dis* [Internet]. 2022 July 4 [cited 2025 Aug 8];13(7):580. Available from: <https://www.nature.com/articles/s41419-022-05034-x>

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